ANDRES LEBBOS

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Heerlen, Netherlands

RESEARCH INTERESTS

Computational Oncology

Clinical Decision Systems

Stochastic Decision Making

Multi-Agent Systems

Network Dynamics

LANGUAGES

Spanish	Native
English	C2
Arabic	C1
Italian	B1
Dutch	A2

PROFILE

AI researcher with a background in Computer Science and Electrical & Computer Engineering, currently pursuing a Master's in Artificial Intelligence. Focused on computational oncology and clinical decision support systems. Experienced in multi-agent systems and network-based modeling for complex systems.

EDUCATION

Maastricht University **Aug. 2024 - Sep. 2026 (expected)**

MSc Artificial Intelligence (Quantum Computing specialization)

GPA **9.09 / 10** Maastricht, Netherlands

Thesis (*in progress*): Modeling angiogenesis and immunotherapy effects on simulated cancer tumor growth

Duke University **June 2016 - May 2019**

BSE Electrical & Computer Engineering, BSc Computer Science

GPA **3.585 / 4** Durham, NC

RESEARCH PROJECTS

Computational Oncology

Maastricht University

Master Thesis **Apr. 2026 - current**

Simulated tumor growth dynamics from microscopy videos using computer vision to extract temporal and topological properties.

Modeled tumor growth dynamics using network flow and swarm intelligence-based simulation of nutrient transport.

Investigated limitations of anti-angiogenic therapies and evaluated the effects of combination treatment strategies.

Multi-Agent Systems & Decision Theory

Maastricht University & *Duke University*

Swarm Intelligence for Search and Rescue **Nov. - Dec. 2025**

Developed a robot simulation using simultaneous localization and mapping (SLAM) for environment exploration and navigation.

Implemented particle swarm optimization based multi-agent coordination for efficient search and rescue tasks in robot swarms.

Strategic Voting with Uncertainty and Manipulation **Nov. 2025**

Analyzed the robustness of voting schemes under strategic vote manipulation and incomplete information, focusing on voter collusion and counter-strategic behavior.

Fianco Game AI **Sep. - Oct. 2025**

Developed an AI agent for a chess-like game, achieving 7th place in a masters level competition against 70 other AI agents.

TECH SKILLS

Programming Languages

Python	Java
MATLAB	C

Machine Learning & Data

PyTorch	TensorFlow
NumPy	SQL

Research Infrastructure

LaTeX	Unix
Hugging Face	GitHub

Software & Systems

AWS	Docker

AWARDS

Ibero-American Mathematical Olympiad

Bronze Medal 2013

Central American & Caribbean Mathematical Olympiad

Bronze Medal 2013

COURSEWORK

Advanced Machine Learning

Multi-Agent Systems

Dynamic Game Theory

Stochastic Decision Making

Quantum Algorithms

Multiple Robot Synchronized Path Planning **Aug. - Dec. 2018**

Designed an efficient, collision-free path-planning algorithm for dozens of synchronized robots in tight passages.

Quantum Algorithms & Stochastic Optimization

Maastricht University

Security of Quartic-Speedup Algorithm **Sep. - Dec. 2025**

Analyzed the security implications of a quartic-speedup quantum algorithm for planted noisy k-XOR problems, evaluating feasibility against classical baselines under simulation constraints

MCTS for Quantum Ansatz Design **Sep. - Dec. 2024**

Developed and evaluated Monte Carlo Tree Search (MCTS) variants for quantum ansatz design, improving efficiency through progressive widening and grouping strategies.

Robotics & Control Systems

Duke University

Sonar Bathymetry Hardware **Jan. - May 2019**

Constructed an economic robot capable of diving up to 10,000 feet to track marine species and map oil reserves for research purposes.

Vehicle Slippage in Unknown Terrain **Aug. - Dec. 2018**

Created a learning algorithm for steering vehicles, reducing slippage through machine learning, feedback, and PID controllers.

WORK EXPERIENCE

Software Reliability Engineer **June 2022 - Feb. 2024**

Art of Problem Solving Bogota DC, Colombia

Maintained and modernized the reliability, resilience, infrastructure, and security of distributed systems serving over a million users.

Led a security committee, reducing system degrading incidents by 20% while improving their resolution and prevention.

Migrated and optimized a 1TB database, achieving 20% performance improvement and 40% cost reduction without any data loss.

Software Engineer **Sep. 2019 - June 2022**

Art of Problem Solving San Diego, CA

Designed and implemented software tools supporting an online learning platform with 10,000 students and 270,000 members.

Mentored and onboarded engineers, contributing to team scaling and technical integration.

Developed an automated grading algorithm with 97% accuracy, including detection of academic dishonesty.